

Mean–Variance Dynamics and Market Efficiency in Silver

A multi-model forecasting framework — weekly silver returns and volatility, 2015–2026

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Q1 · WEAK FORM

NO

Are weekly returns forecastable from their own past?

AIC picks ARIMA(0,0,0), the drift itself; the spectral test can’t reject the martingale.

Q2 · SEMI-STRONG

NO

Does public information help, linear or nonlinear?

0 of 16 model selections beat the drift out of sample (smallest DM $p = 0.43$).

Q3 · THE VARIANCE

YES

Is weekly realised volatility forecastable?

Every model beats the naïve floor; HAR alone cuts its mean-squared error by a third.

Q4 · SENTIMENT

VARIANCE ONLY

Does Reddit & news sentiment add anything?

Nothing to the mean; a small but robust gain to volatility (QLIKE $p < 0.01$).

The mean is a *martingale*; the variance is *not*.

Predictable risk alongside an unpredictable return is exactly what an efficient market permits — one statement, not two studies.

01 One series, two moments

591 Friday-to-Friday weeks · target $r_t = \log P_t - \log P_{t-1}$

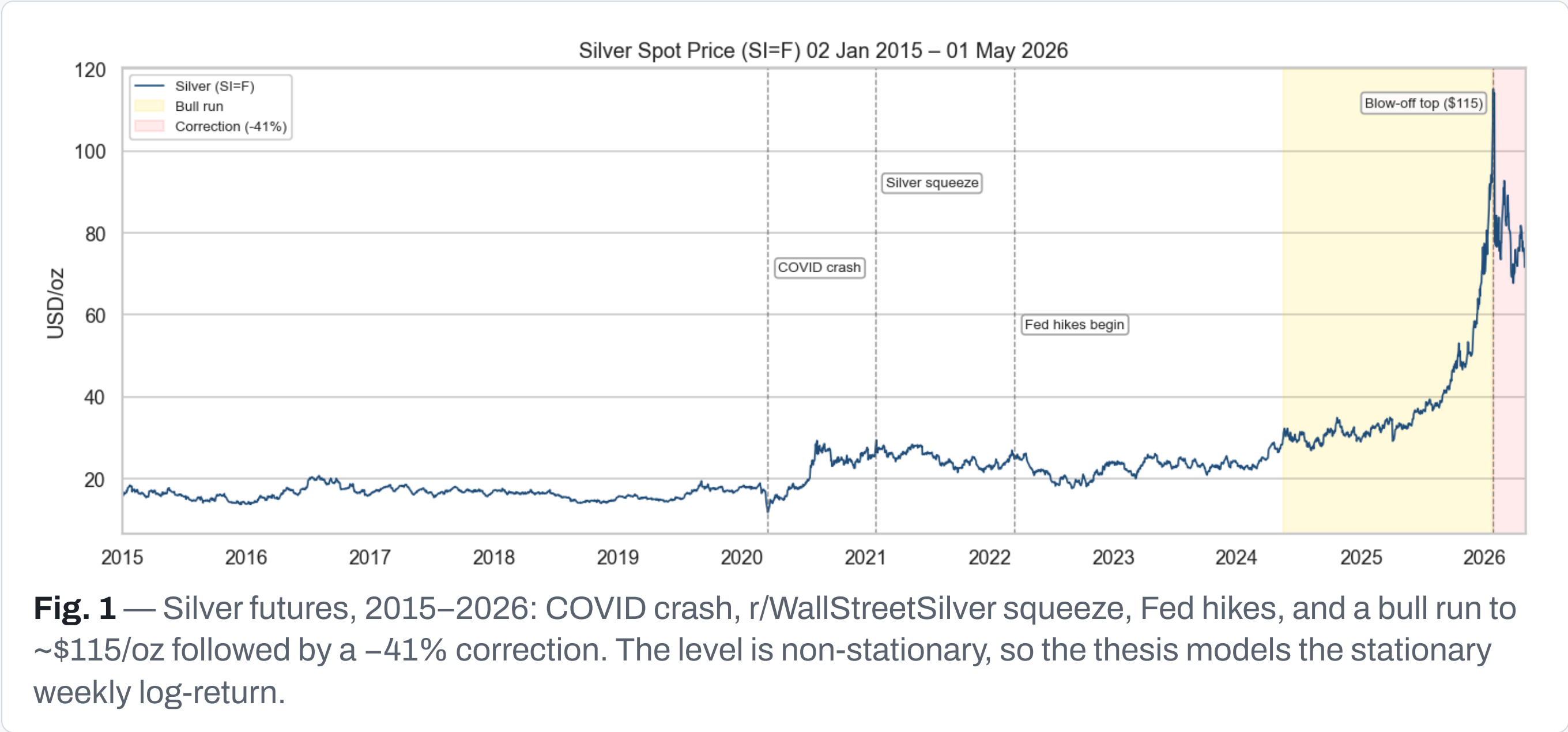


Fig. 1 — Silver futures, 2015–2026: COVID crash, r/WallStreetSilver squeeze, Fed hikes, and a bull run to ~\$115/oz followed by a –41% correction. The level is non-stationary, so the thesis models the stationary weekly log-return.

Silver lives a double life: an industrial input (~half of demand: photovoltaics, electronics), a monetary hedge, and an episodic meme trade (spot \$25→\$30 in days, Jan 2021). An unusually rich laboratory for the oldest question in empirical finance: *how much of a price can be forecast?*

Returns: almost no autocorrelation, so a linear model has nothing to grip.
Squared returns: strongly autocorrelated: volatile weeks cluster.
The structure is not gone; it lives in the second moment.

The weekly horizon is where the information aligns: positioning is reported weekly, macro monthly — coarse enough to average out the microstructure noise that dominates daily returns, yet fine enough to leave ~600 weeks to train and honestly test the ML models.

02 Six source families, one weekly frame

every feature lagged before use: no look-ahead

Market prices
Yahoo · daily: silver + 6 cross-assets

Rates & inflation
FRED · daily: real yield, breakevens, claims

Macro releases
FRED · monthly: point-in-time rule

Positioning
CFTC COT · weekly: funds vs hedgers

Sentiment
Reddit · news · Trends: RoBERTa + FinBERT

China PMI
FRED · monthly: industrial demand

591 weeks → Train 2015–21 365 wk Val '22 Test 2023–26 174 wk

03 The mean: nothing beats the drift

ARIMA(X) · VAR · MIDAS · RF · XGBoost · LSTM, all vs random-walk-with-drift, walk-forward

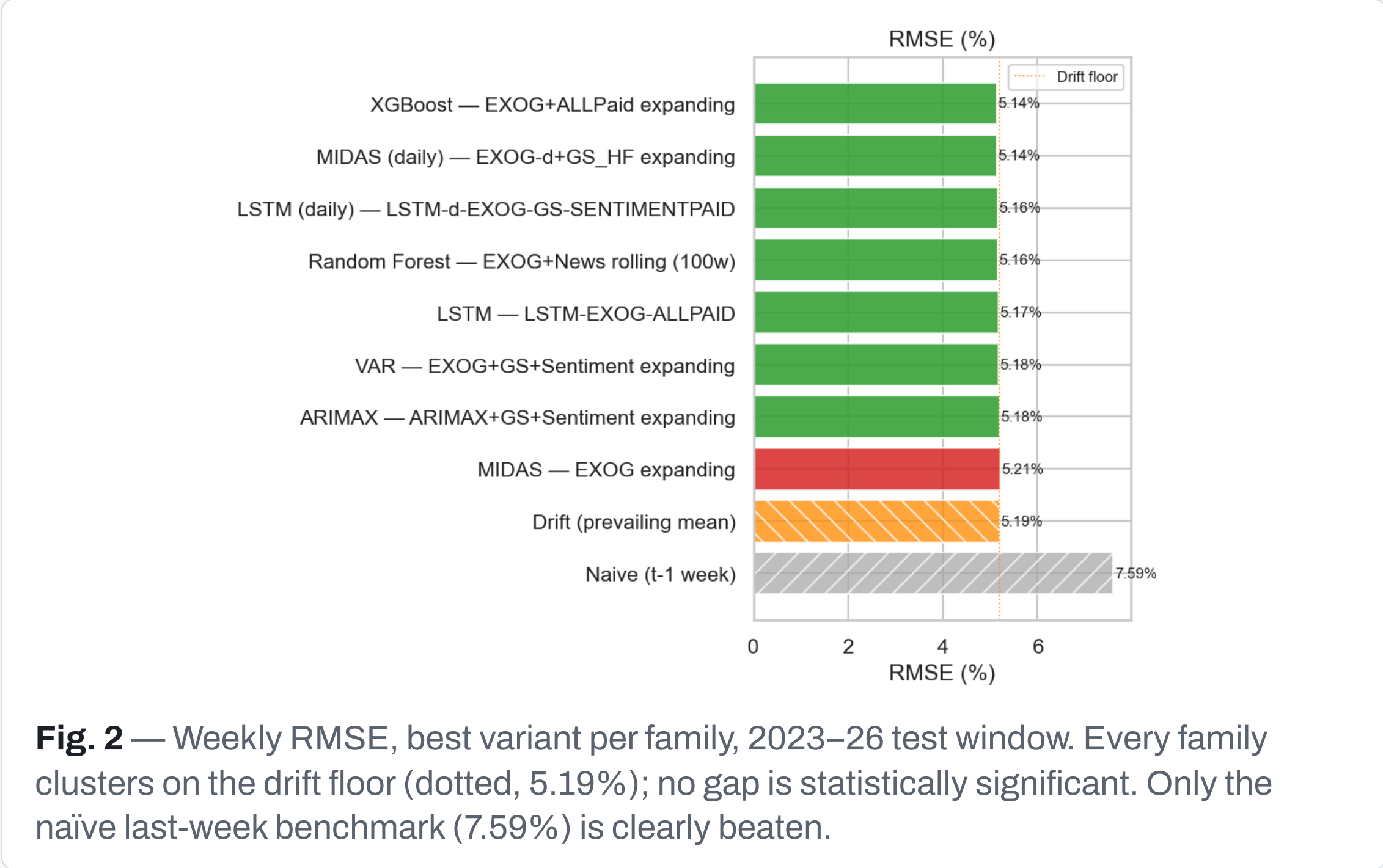


Fig. 2 — Weekly RMSE, best variant per family, 2023–26 test window. Every family clusters on the drift floor (dotted, 5.19%); no gap is statistically significant. Only the naïve last-week benchmark (7.59%) is clearly beaten.

0 / 16

family selections beat the drift on a Diebold–Mariano test

0.43

smallest DM p -value across every family and loss function

A directional footnote. Linear models time the sign (Pesaran–Timmermann $p \approx 0.005$); WDA peaks at 0.76–0.79 (ARIMAX '23, VAR '25) — but always-up alone scores 0.757 in the 2025 bull, and the signal is magnitude-worthless (negative R^2_{OS}), fading in the nonlinear families. An adaptive-markets curiosity, not a rejection of efficiency.

Out-of-sample R^2 vs drift, by sub-period (%, RMSE-best variant per family)				
	'23 choppy	'24 bull start	'25 bull run	'26 YTD
ARIMAX	+5.4	+7.3	–3.3	–2.5
VAR	+5.2	+6.7	–3.9	–1.4
Random Forest	–4.7	–3.4	+3.2	+3.8
XGBoost	+0.3	+5.8	+2.5	+0.5
LSTM	–0.9	+4.1	–0.3	+1.0
MIDAS	+0.5	+2.0	–3.7	–0.1
MIDAS (daily)	–0.7	+12.4	–8.0	+5.5
LSTM (daily)	+1.7	+2.1	–1.0	+2.2

Apparent skill is regime-dependent: ARIMAX & VAR shine in 2023–24 and invert in the 2025 bull. Only **XGBoost** stays positive in all four periods (+1.8% full-period R^2_{OS}), never reaching DM significance.

04 The variance: strongly forecastable

weekly realised volatility · GARCH · HAR · tree ensembles, vs naïve RV_{t-1}

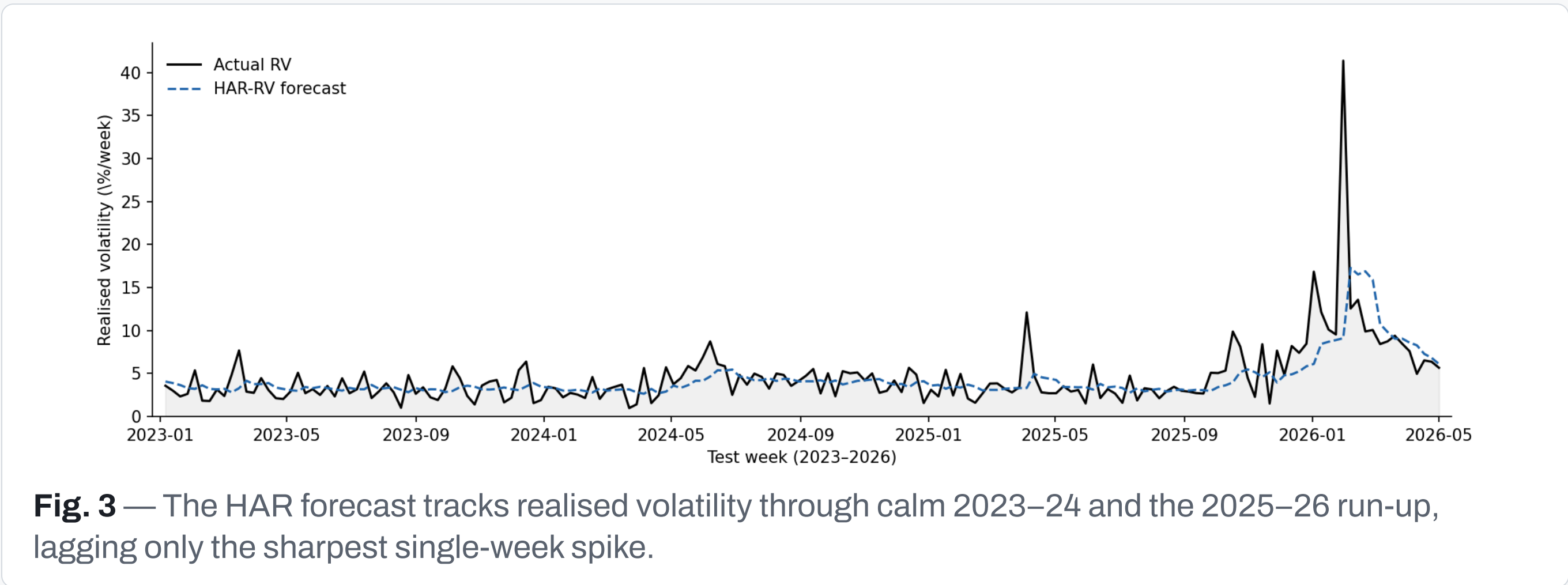


Fig. 3 — The HAR forecast tracks realised volatility through calm 2023–24 and the 2025–26 run-up, lagging only the sharpest single-week spike.

–1/3 MSE

vs the naïve floor: $R^2_{OS} = 0.29\text{--}0.32$, all models

71%

of weekly volatility rises & falls called correctly (HAR)

$p < 0.01$

QLIKE Diebold–Mariano vs naïve, every model

Forecast error, RV points (RMSE), lower is better

XGBoost (HAR lags)

3.13

Random Forest (HAR lags)

3.15

HAR-RV

3.20

GARCH(1,1)

3.28

Naïve (RV_{t-1})

3.99

Robust across model classes:

GARCH, HAR, and trees-on-HAR all beat naïve (QLIKE-DM $p < 0.01$), not one lucky spec. Cross-asset spillover **hurts** (HAR+EXOG loses to bare HAR, $p \approx 0.03$): silver's forecastable variance is its own, not gold's or copper's. Collapsing HAR to one lag degrades it: the signal is multi-horizon persistence.

05 Attention prices risk, not direction

Does a text signal improve HAR? QLIKE DM statistic: left of zero improves

Reddit sentiment

–3.36 ***

Paid-news attention

–3.33 ***

Reddit both

–3.01 **

Reddit attention

–2.60 **

Google Trends

–0.01 (tie)

Paid-news sentiment

+1.44

Cross-asset spillover

+2.23 * worse

Linear HAR runs, 174 common weeks. **How much is written** generally beats **how it reads**: attention and tone-intensity help; polarity alone does not. The same block never helps the mean.

06 Does it matter? Ask a risk manager

One-week 5% Value-at-Risk backtest breach should sit near 5% · pinball $\times 10^4$, lower is better

VOLATILITY FORECAST	BREACH	PINBALL
Naïve (RV_{t-1})	8.0%	66.3
GARCH(1,1)	3.4%	58.5
HAR	6.3%	59.6
HAR + Reddit attention	5.7%	59.6

HAR-based VaR beats the naïve floor at $p < 0.001$: the naïve VaR over-breaches (8% hits at the 5% tail) and overshoots after every spike. GARCH is the most conservative; Reddit attention nudges HAR's breach rate closest to the 5% target (5.7%). We build the sentiment story on HAR, where the signal is cleanest and QLIKE-significant. Forecastable volatility is the difference between a usable and an unusable tail-risk number: position sizing, options, and risk budgets consume the *variance*, not the mean.